Selective quadratic isolation and sparse target padding

Explicit source-encoding certificate

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1 Selective quadratic isolation with sparse target padding

This section proves the new source encoding, independently of the number-theoretic finalizers. Its only finite input is the explicit tuple in `code_v1.json` and the 22 source residuals identified below. The proof does not assume a full Sidon (B_2) property, disjoint source intervals, or any unverified output of the search program.

1.1 The exact source and coefficient convention

Let $r_1, \dots, r_{22}$ be the quadratic residuals of the original package31 positive-variable source. Every original variable X_i is replaced by $x_i + 1$, where $x_i \in \mathbb{N} = \{0, 1, \dots\}$. We use the following coordinate order:

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma, \mathsf{N}, \mathsf{E}, \mathsf{P}, c, a,$$
$$X, D, C_2, L, F_2, V, g, I, B_{\mathrm{aux}}, \Delta.$$

Here the displayed names label source coordinates, not the variables of a final nine- or ten-variable polynomial. In this section the coefficients may first be regarded as elements of $\mathbb{Z}[\beta, K_0, M_C, M_F]$; the final construction must substitute its specified, fixed compiler tuple. These four symbols are not additional Diophantine unknowns. The isolation identities below hold under every integer specialization of the ports.

The exact source bytes are `upstream_parent_quartic31.json`, fetched from repository commit

`124c07c0b77a9e20e84676514d8b1ff78536ea3f`.

The accompanying `SOURCE_PINS.json` records repository paths, SHA-256 hashes and Git blob hashes. The selected source is the original positive-transport version, not a silent substitution of package45's transport variant. After shifting coordinates, package45's row 15 is precisely the selected row 15 minus $\beta(x_J + 1)$. Thus its weighted constant coefficient differs by -2β, and its weighted x_J coefficient by $-\beta$; the other weighted coefficients agree. All 136 supports are the same. The checker verifies this exact difference.

1.2 A support-selective isolation theorem

For n coordinates write

$$r_k(x) = \sum_{\alpha \in S_k} p_{k,\alpha} x^\alpha, \qquad |\alpha| \le 2.$$

Choose distinct positive integer weights $w_1, \dots, w_n$ and set $w_0 = 0$. Pad each multi-index α with $2 - |\alpha|$ copies of index 0 to obtain an unordered pair. This is a bijection between multi-indices of degree at most two and the $\binom{n+2}{2}$ unordered pairs from $\{0, \dots, n\}$. Put $e(\alpha) = \sum_i \alpha_i w_i$. Choose distinct targets $q_k \in \mathbb{N}$ satisfying

$$q_k - e(\alpha) \ge 0 \qquad (\alpha \in S_k), \tag{1}$$
$$q_k - e(\alpha) + e(\gamma) = q_j \implies k = j \text{ and } \alpha = \gamma \qquad (\alpha \in S_k,\ |\gamma| \le 2). \tag{2}$$

Only the first monomial in this condition is required to occur in a residual. Distinct absent monomials are allowed to have equal weights. Define

$$f_\alpha = (2-|\alpha|)!\prod_i \alpha_i!, \qquad c_{k,\alpha} = f_\alpha p_{k,\alpha}, \qquad D(t) = \sum_{k,\alpha\in S_k} c_{k,\alpha} t^{q_k - e(\alpha)},$$

and $C_x(t) = (1+\sum_i x_i t^{w_i})^2$. Then, identically in all the x_i and all coefficient ports,

$$[t^{q_j}]C_x(t)D(t) = 2r_j(x). \tag{3}$$

Indeed, the contribution from monomial x^γ in the square has multinomial coefficient $2/f_\gamma$. Condition (2) permits it to reach target q_j only when $k=j$ and $\gamma=\alpha$. Its product with $f_\alpha p_{j,\alpha}$ is $2p_{j,\alpha}$. Summing proves the identity, including all signs. This argument is a polynomial identity, not a test on bounded witnesses.

The coefficient bound does not acquire a factor 22. The reversed exponents $q_k - e(\alpha)$ are injective. To see this, suppose $q_k - e(\alpha) = q_j - e(\gamma)$ with $\alpha \in S_k$ and $\gamma \in S_j$. Adding $e(\gamma)$ makes the left side reach q_j; (2) forces $k=j$ and $\alpha=\gamma$. Consequently, after fixing the integer ports, if $A = \max_{k,\alpha}|c_{k,\alpha}|$, each coefficient of the entire polynomial D has magnitude at most A. For $x_i \geq 0$,

$$\big|[t^s]C_x(t)D(t)\big| \leq A\left(1+\sum_i x_i\right)^2. \tag{4}$$

For a fixed coefficient of the product, sum over the nonnegative coefficients of C_x and use the bound A for the single corresponding coefficient of D. Their sum is $C_x(1)$. Collisions between absent monomials in C_x, or overlaps between source intervals, do not change this argument.

Leading coefficient and positivity. Let $M = \max_i w_i$ and $Q = \max_k q_k$. In the explicit certificate the unique row at target Q is original row 5. Its shifted polynomial is

$$r_5(x) = 1 + 2x_E - x_a + x_E x_X - x_a x_X.$$

It has constant coefficient 1. Positivity of every input weight implies that the coefficient of t^Q in D is exactly 2, and no term has higher degree. If $B > A+1$, then

$$\left|\sum_{s<Q}[t^s]D(t)B^s\right| \leq A\frac{B^Q-1}{B-1} < B^Q, \qquad D(B) > 0.$$

Also $D(B) < 2AB^Q$ for $B \geq 2$. Row 5 shows that the all-zero source vector is not a common zero. These facts hold independently of the values of the fixed ports.

1.3 The explicit frozen certificate

The weight vector, including its padding coordinate, is

$$\begin{aligned}(w_0,\ldots,w_{31}) = (&0,9,25,23,139,35,151,168,21,91,181,94,52,150,180,7,\\ &135,38,88,183,106,178,136,20,138,8,64,105,154,37,137,120).\end{aligned}$$

The row order and corresponding targets are

k	16	2	9	14	21	7	15	22	13	3	19
q_k	70	86	451	685	1050	1296	1530	1874	2212	2358	2696
k	20	1	10	17	8	4	6	11	18	12	5
q_k	2780	2792	3157	3290	3634	3929	4238	4384	4527	4865	5190

Thus

$$M = 183, \qquad Q = 5190, \qquad H = Q + 2M = 5556, \qquad (M+1) + (H+1) = 5741.$$

There are 71 marked padded pairs: the 32 constant/linear pairs and 39 genuine quadratic pairs. The 22 row term counts, in original row order, are

$$(4, 5, 5, 6, 5, 8, 6, 8, 8, 4, 3, 6, 9, 17, 10, 3, 6, 6, 4, 3, 4, 6),$$

with sum 136. Of all 528 padded pairs, only 267 different integer pair sums occur. For example, $w_1 + w_3 = 9 + 23 = 25 + 7 = w_2 + w_{15}$; both pairs are absent from the source. Therefore this is genuinely not a full B_2 set. The overlapping intervals, including targets only 12 apart, are intentional.

The byte-level canonical identifier for `code_v1.json` is

`53cfdccbb252fd3f6ebe87ae0bb9dad8b2fdfc8c6168f668029add30ed339e25`.

No seed, random choice, optimizer or search executable remains in the mathematical specification. A deterministic finite tuple is part of the equation, rather than an existential claim that some suitable tuple exists.

1.4 Sparse target padding: one bit absorbs incoming borrow

The same selective code permits a smaller source wrapper than dense centering. Let $B \geq 4$ be a power of two and let

$$F(t) = \sum_{s=0}^{H} a_s t^s \in \mathbb{Z}[t], \qquad |a_s| < B/4, \qquad F(B) > 0.$$

For a target set $\mathcal{Q} \subseteq \{0, \ldots, H\}$ put

$$A_{\mathcal{Q}} = \sum_{q \in \mathcal{Q}} B^q, \qquad J = 2F(B) + A_{\mathcal{Q}}.$$

Then

$$\tau_2(J, (B-2)A_{\mathcal{Q}}) = 0 \quad \Longleftrightarrow \quad a_q = 0 \quad \text{for all } q \in \mathcal{Q}, \tag{5}$$

where $\tau_2(u, v) = 0$ means that binary addition of the nonnegative integers u, v is carry-free.

For completeness, write $\delta_s = 1$ at a target and 0 elsewhere, and form the signed carry recurrence

$$c_0 = 0, \quad c_{s+1} = \left\lfloor \frac{2a_s + \delta_s + c_s}{B} \right\rfloor, \quad d_s = 2a_s + \delta_s + c_s - Bc_{s+1}.$$

Induction gives $c_s \in \{-1, 0\}$ and $0 \leq d_s < B$: the numerator lies strictly between $-B$ and B. Because $J > 0$, telescoping forces the final carry $c_{H+1} = 0$; otherwise $J = \sum_{s=0}^{H} d_s B^s - B^{H+1} < 0$. Thus the d_s are the actual base-B digits of J.

At a target the mask digit is $B - 2$, whose binary digits forbid all bits except the least significant bit. Hence carry-freeness requires $d_s \in \{0, 1\}$. If the outgoing carry were -1, the numerator would be $d_s - B \leq 1 - B$, contradicting $2a_s + 1 + c_s > -B/2 - 1 \geq 1 - B$. The outgoing carry is therefore zero. Now $2a_s = d_s - 1 - c_s$, and the only even integer allowed by $d_s \in \{0, 1\}$ and $c_s \in \{-1, 0\}$ is 0. Conversely, $a_s = 0$ gives $d_s = 1 + c_s \in \{0, 1\}$ and outgoing carry zero. This proves (5); in particular, an incoming borrow cannot produce a false zero test. No test is imposed on an absent target.

1.5 Source wrapper and size estimates before power-of-two decoding

Choose an integer $b \geq 1$ and an integer base $B > 4096Ab^2$. This assumption is a numerical coefficient majorant, not yet a power-of-two assumption. Define

$$\begin{aligned} \ell &= \sum_{i=1}^{31} B^{w_i}, & A_{\mathcal{Q}} &= \sum_{k=1}^{22} B^{q_k}, \\ N_0 &= B^{M+1}, & N_1 &= B^{H+1}, \\ N_{\mathrm{src}} &= N_0 N_1 = B^{5741}, & \mathcal{C} &= bB^M, \\ M_0 &= N_0 - 1 - (b-1)\ell, & J(g) &= 2(1+g)^2 D(B) + A_{\mathcal{Q}}, \\ S(g) &= g + N_0 J(g), & T &= M_0 + N_0(B-2)A_{\mathcal{Q}}. \end{aligned}$$

For every integer $0 \leq g < 2\mathcal{C}$, all of these expressions needed by the no-carry test are nonnegative, with $J(g), T > 0$, and

$$S(g) + T < N_{\mathrm{src}}. \tag{6}$$

Indeed, $B > 2b$ implies $g < N_0$, and the base-B digits of M_0 are $B-b$ at the distinct marked input positions and $B-1$ elsewhere, so $0 \leq M_0 < N_0$. As g is integral, $1 + g \leq 2bB^M$. Consequently

$$0 < 2(1+g)^2 D(B) < 16Ab^2 B^H < N_1/256.$$

Furthermore, using $M \geq 1$,

$$(B-1)A_{\mathcal{Q}} \leq B^{Q+1} - 1 < N_1/B^2.$$

It follows that

$$S(g) + T < N_0 \left(2 + N_1/256 + N_1/B^2\right) < N_0 N_1.$$

The last inequality follows already from $B \geq 4$ and $N_1 \geq 4$. These estimates hold before any bridge has deduced that b is a power of two. They therefore do not introduce circular reasoning into that bridge. The displayed N_{src} has no factor 4; the old dense-padding wrapper $4B^{5741}$ is not the wrapper used here.

If now b and B are powers of two, binary carry-freeness of g and M_0 is equivalent to a unique representation

$$g = \sum_{i=1}^{31} x_i B^{w_i}, \qquad 0 \leq x_i < b.$$

The mask $B - b$ allows precisely the lowest $\log_2 b$ bits in each marked cell and no bits in an unmarked cell. For this representation, $(1+g)^2 D(B) = C_x(B) D(B)$. By (4),

$$|a_s| \leq A\big(1 + 31(b-1)\big)^2 \leq A(31b)^2 < B/4.$$

The low and high words split at the power of two N_0. Combining the low-word mask, (5) and (3) proves

$$\tau_2(S(g), T) = 0 \quad \Longleftrightarrow \quad \begin{cases} g = \sum_i x_i B^{w_i}, \quad 0 \leq x_i < b, \\ r_1(x) = \cdots = r_{22}(x) = 0. \end{cases}$$

A common source zero is nonzero because $r_5(0) = 1$. Choosing b from any cofinal allowed power-of-two family larger than every x_i gives $B \leq g < \mathcal{C}$, since all weights are positive and distinct. Thus the usual cofinal source-witness requirement is preserved. The arithmetic finalizers must still prove their own bridge hypotheses, fixed coefficient specialization and final variable-domain equivalences.

1.6 Independent finite verification and limits

The standard-library program `verify_selective.py` performs the following checks without executing any upstream code or search binary:

1. Check source SHA-256 and Git blob hashes against the current pinned repository receipts, and the frozen code's exact byte hash.

2. Parse all 22 parent residual formulas, including affine abbreviations, in an independently implemented sparse polynomial ring; compare them with the supplied sparse terms.

3. Apply the 31 binomial shifts, verify every weighted coefficient in `source_support.json`, and compare all 136 terms to package45 with precisely the two row-15 corrections described above.

4. Enumerate all $136 \cdot 528 = 71\,808$ source-term/padded-pair interactions. Exactly 136 land at marked targets, each at its intended row and monomial. Reconstruct the 22 formal target polynomials and verify that each is twice its residual.

5. Check all reversed exponents are nonnegative and distinct, the leading coefficient is 2, and the displayed maxima are exact.

The replay command is `python3 verify_selective.py`; it also passes with Python's `-O` switch, because validity checks do not use removable assertions. The finite certificate supplies the complete finite premise of the isolation proof. It does not purport to prove the inherited machine compiler or the Sun/Jones number-theoretic theorems.

The maximum target decreases from the package45 value 35178 to 5190, and the maximum input weight from 799 to 183. These are source exponent spans, not final polynomial degrees or operation counts; those metrics must be computed from each compiled finalizer. The tuple is not claimed globally optimal. The construction gives no new proof of Con(ZFC), and no assertion that the consistency question has been decided. Its contribution is an explicit, rigorously checkable support-selective code and a sparse padding lemma that can be composed with the separately stated consistency-to-source theorem.